

TrES-2b

R-2545

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_110822 · created 2026-08-26T00:26:35+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455795.77201 +/- 0.0001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.019
Transit depth (Rp/Rs)^2	2.19 +/- 0.55 [%]
Orbital Inclination (inc)	83.25 +/- 0.62
Airmass coefficient 1 (a1)	1.0494 +/- 0.0092
Airmass coefficient 2 (a2)	-0.04 +/- 0.011
Scatter in the residuals of the lightcurve fit is	0.88 %
Best Comparison Star	#1 - [567, 228]
Optimal Aperture	3.56
Optimal Annulus	8.12
Transit Duration (day)	0.066 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.019 vs 0.1254 (deviation 1.19σ)
Duration fit vs published	0.066 d vs 0.0746 d (deviation 0.61σ)
Mid-time O–C	0.2 min
Depth SNR	7.8
Residual scatter	0.88 %

Light curve

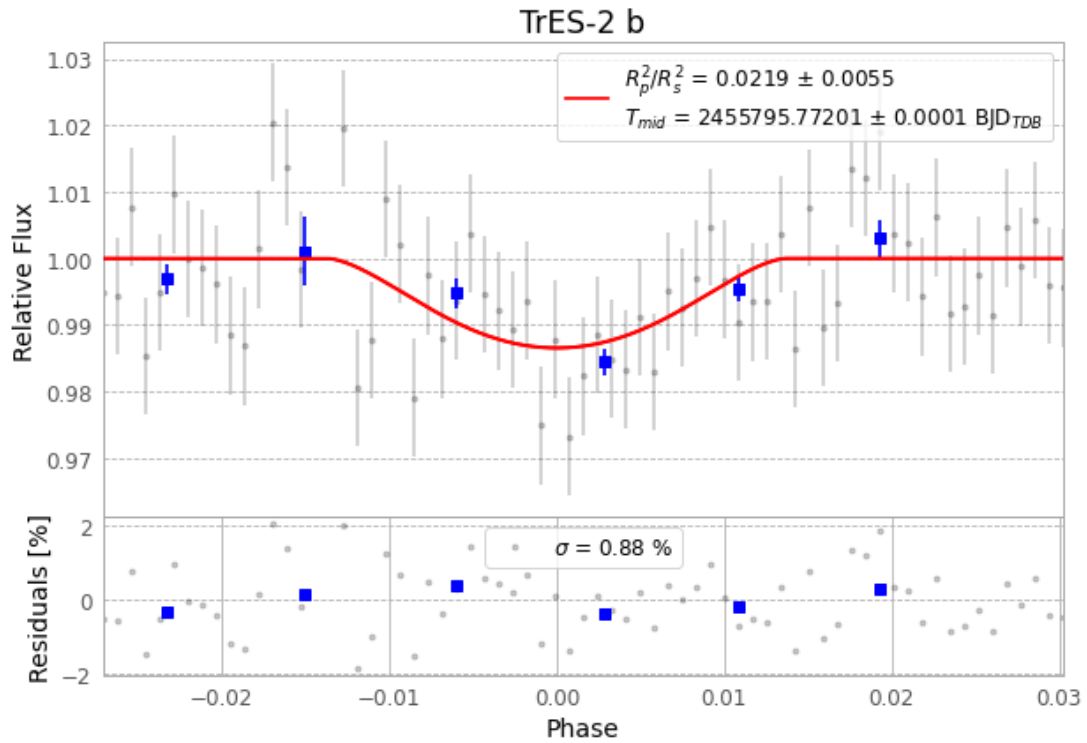


Figure 1 — FinalLightCurve_TrES-2 b_2011-08-21.png (SHA-256 7e4f4d34ab61404a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [561, 293], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0d33207935a88a9b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

69 FITS frames (45 MB), TRES-2110822045112.FITS → TRES-2110822081512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-110822.log (e43de5bf7f0f...), FinalLightCurve_TrES-2 b_2011-08-21.png (7e4f4d34ab61...), FinalLightCurve_TrES-2 b_2011-08-21.csv (08652ac9e3b9...)

Compute resources

Wall time 113.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 00:26Z.